THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 5–8: **Parallel and Intersecting Lines, Number Play, A Tale of Three Intersecting Lines, Working with Fractions** | Class 7 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** 66°	Their sum is 90°.
2	**(b)** 77	Sum $= 11(a+b)$.
3	**(c)** 7 cm	The third side lies between the difference and the sum of the other two.
4	**(d)** 12	Number of pieces $= n \div$ piece length.
5	**(d)** 60°	The angles add up to 180° in 3 equal parts.
6	**(d)** 63	Difference $= 9(a-b)$.
7	**(c)** 30	Add the angles: $(3+k)x = 180$.
8	**(b)** 15	Scaling by a fraction is multiplication.
9	**(d)** 115°	Interior angles on the same side are supplementary.
10	**(a)** 5	List the terms and count the steps.
11	**(d)** 30	Exterior angle = sum of interior opposite angles.
12	**(a)** 8	Use the meaning of multiplying and dividing fractions.
13	**(c)** parallel	Use the tests for parallel lines.
14	**(b)** 2	Digit sum must be a multiple of 9.
15	**(c)** QR	Use the idea of corresponding parts.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	The angles are y and $2y$, so $3y = 180$; $y = 60$ (1); The angles are 60° and 120° (1).	1+1
17	$10a + b + 10b + a = 11a + 11b$; $= 11(a+b)$ (1); For $a = 3, b = 5$: $35 + 53 = 88 = 11 \times 8$ (1).	1+1
18	Third side $> 6 - 10$ in absolute value $= 4$; Third side $< 6 + 10 = 16$ (1); Least whole number 5 cm; Greatest whole number 15 cm (1).	1+1
19	Count $= \frac{3}{4} \div \frac{1}{8}$; $= \frac{3}{4} \times \frac{8}{1} = \frac{24}{4}$ (1); Answer 6 (1).	1+1
20	Compare 75° and 75° (1); Equal corresponding angles, so the lines are parallel (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Fourth angle $= 360^\circ - (95 + 85 + 110)^\circ = 70^\circ$ (1); Kinds: 95° obtuse, 85° acute, 110° obtuse, 70° acute (2).	1+2
22	Step 1: $9652 - 2569 = 7083$ (1); Step 2: $8730 - 0378 = 8352$ (1); After two steps the number is 8352 (1).	1+1+1
23	$\angle B = \angle C$ (equal sides); $(x+10) + 2(2x-5) = 180$, so $5x = 180$ (1); $x = 36$ (1); $\angle A = 46^\circ$, $\angle B = \angle C = 67^\circ$ (1).	1+1+1
24	Area $= \frac{3}{4} \times \frac{2}{5} = \frac{3}{10}$ m^2 (1); Smaller side squared $= \frac{4}{25}$ m^2 (1); Difference $= \frac{7}{50}$ m^2 (1).	1+1+1
	OR: For parallel lines these angles must be equal (1); Here 65° and 65° are equal (1); The lines are parallel (1).	1+1+1
25	$57 + 75 = 132$; $132 + 231 = 363$ (1); Number of steps $= 2$ (1); Palindrome: 363 (1).	1+1+1
	OR: Two sides and the included angle are equal, so SAS; $\triangle ABC \cong \triangle PQR$, hence $PR = AC = 9$ cm (1); Perimeter of PQR $= 5 + 7 + 9$ (1); $= 21$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Equal to 50°: the angle itself, its vertically opposite angle, its corresponding angle and its alternate interior angle, that is 4 angles (1); Every other angle forms a linear pair with one of these, so it measures 130° (4 angles) (2); Check: $50 + 130 = 180$ (2).	1+2+2
	OR: Reverse $= 237$ (1); Difference $= 732 - 237 = 495$ (1); $= 99 \times 5$ (1); In general $100a + c - 100c - a = 99(a-c)$ (1); So the difference is always a multiple of 99 (1).	1+1+1+1+1

27	$4x + 20 = 180$, so $x = 40$ (1); Angles 50°, 30°, 100° (1); Classification: obtuse-angled (1); Exterior angles 130°, 150°, 80° (1); Their sum $= 360^\circ = 360^\circ$ (1).	1+1+1+1+1
	OR: First part $= \frac{1}{4} \times 120 = 30$ km (1); Remaining $= 120 - 30 = 90$ km (1); Second part $= \frac{2}{3} \times 90 = 60$ km (1); Left $= 90 - 60$ (1); $= 30$ km (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 5–8: **Parallel and Intersecting Lines, Number Play, A Tale of Three Intersecting Lines, Working with Fractions** | Class 7 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** 6	Convert to an improper fraction and multiply.
2	**(b)** 28°	Vertically opposite angles are equal.
3	**(d)** 55	Triangular number $\frac{n(n+1)}{2}$.
4	**(a)** $\angle A$	The included angle lies between the two given sides.
5	**(d)** $\frac{2}{3}$	Write as an improper fraction, then invert.
6	**(b)** acute	Classify by the measure: acute $< 90^\circ$, right $= 90^\circ$, obtuse between 90° and 180°, reflex above 180°.
7	**(d)** 0 or 5	Recall the divisibility test.
8	**(b)** $50^\circ, 60^\circ, 70^\circ$	The three angles must add up to 180°.
9	**(d)** 2	Multiply by the reciprocal of the divisor.
10	**(a)** 90°	The angles around a point add up to 360°.
11	**(a)** 40	Pairs around the middle cancel out.
12	**(a)** $\angle B = \angle C$	Equal angles and equal sides go together in an isosceles triangle.
13	**(a)** $4\frac{1}{2}$	Multiply the quantity by the number of batches.
14	**(c)** 53°	Their sum is 90°.
15	**(c)** 7531	Write the digits in decreasing order.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Known digit sum $= 13$; Total $= (13 + *)$ must be a multiple of 9 (1); $* = 5$; Check: 4527 has digit sum 18 (1).	1+1
17	$6x + 18 = 180$; $x = 27$ (1); Angles 64°, 77°, 39° (1).	1+1
18	$\frac{3}{4}$ of 48$= 36$; $\frac{2}{3} \times 36$ (1); $= 24$ kg (1).	1+1
19	Let the angle be y; then $y + (y + 20) = 180$; $2y = 160$ (1); $y = 80^\circ$ (1).	1+1
20	Middle number $= 110 \div 5 = 22$; The numbers are $18, 20, 22, 24, 26$ (1); Check: sum $= 110$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Improper form $\frac{5}{2}$; $\frac{5}{2} \times \frac{4}{3}$ (1); $= \frac{20}{6}$ (1); Answer $3\frac{1}{3}$ (1).	1+1+1
22	The sum is 90° and the number of parts is 3 (1); One part $= 30^\circ$ (1); The angles are 30° and 60° (1).	1+1+1
23	Sum of three consecutive numbers $= 3\times$ middle (1); $63 \div 3 = 21$ is a whole number, so yes (1); The numbers are $20, 21, 22$ (1).	1+1+1
24	$\angle A + \angle B = 120^\circ$; One part $= 120 \div 4 = 30^\circ$ (1); $\angle A = 30^\circ$, $\angle B = 90^\circ$ (1); $\angle C = 180^\circ - 120^\circ = 60^\circ$ (1).	1+1+1
	OR: Cloth for 8 shirts $= 8 \times \frac{5}{4}$; $= 10$ m (1); Shirts from 10\frac{5}{8} m $= 8\frac{1}{2}$ (1); Full shirts $= 8$ (1).	1+1+1
25	Fourth angle $= 360^\circ - (120 + 70 + 60)^\circ = 110^\circ$ (1); Kinds: 120° obtuse, 70° acute, 60° acute, 110° obtuse (2).	1+2
	OR: Step 1: $7521 - 1257 = 6264$ (1); Step 2: $6642 - 2466 = 4176$ (1); After two steps the number is 4176 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Let the supplement be y; then the angle is $2y$ and $2y + y = 180$ (1); $y = 60$ (2); The angle is 120° and its supplement is 60° (2).	1+2+2
	OR: Sequence: $17, 52, 26, 13, 40, 20, 10, 5, 16, 8, 4, 2, 1$ (1); Halvings $= 9$ (1); Triple-and-add-1 steps $= 3$ (1); Total steps $= 12$ (2).	1+1+1+2

27	(i) Between 5 cm and 19 cm (1); (ii) Whole numbers 6 to 18: 13 values (1); (iii) Smallest third side 6: perimeter 25 cm (1); Largest third side 18: perimeter 37 cm (1); State the answers with units (1).	1+1+1+1+1
	OR: Area $= \frac{4}{5} \times \frac{5}{6} = \frac{2}{3}$ (1); Vegetable part $= \frac{3}{4} \times \frac{2}{3} = \frac{1}{2}$ (1); Flower part $= \frac{2}{3} - \frac{1}{2}$ (1); $= \frac{1}{6}$ (1); State the three areas with units (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 5–8: **Parallel and Intersecting Lines, Number Play, A Tale of Three Intersecting Lines, Working with Fractions** | Class 7 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** not necessarily congruent	AAA does not guarantee congruence.
2	**(a)** 14	Number of pieces $= n \div$ piece length.
3	**(a)** 36°	The angles add up to 180° in 5 equal parts.
4	**(b)** 99	Sum $= 11(a+b)$.
5	**(c)** 10 cm	The third side lies between the difference and the sum of the other two.
6	**(b)** 14	Scaling by a fraction is multiplication.
7	**(b)** 100°	Interior angles on the same side are supplementary.
8	**(c)** 36	Difference $= 9(a-b)$.
9	**(b)** 20	Add the angles: $(3+k)x = 180$.
10	**(b)** 2	Use the meaning of multiplying and dividing fractions.
11	**(d)** parallel	Use the tests for parallel lines.
12	**(d)** 8	List the terms and count the steps.
13	**(a)** 35	Exterior angle = sum of interior opposite angles.
14	**(d)** 14	Convert to an improper fraction and multiply.
15	**(c)** 64°	Vertically opposite angles are equal.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Exterior angle $= \angle A + \angle B$; $5x - 10 = 3x + 50$, so $2x = 60$ (1); $x = 30$; $\angle ACD = 140^\circ$ (1).	1+1
17	$x \times \frac{2}{5} = \frac{4}{15}$; $x = \frac{4}{15} \div \frac{2}{5}$ (1); $x = \frac{2}{3}$ (1).	1+1
18	The angles are y and $3y$, so $4y = 180$; $y = 45$ (1); The angles are 45° and 135° (1).	1+1
19	$10a + b + 10b + a = 11a + 11b$; $= 11(a+b)$ (1); For $a = 4, b = 7$: $47 + 74 = 121 = 11 \times 11$ (1).	1+1
20	Third side $> 4 - 13$ in absolute value $= 9$; Third side $< 4 + 13 = 17$ (1); Least whole number 10 cm; Greatest whole number 16 cm (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	$\angle B = \angle C$ (equal sides); $(x+20) + 2(2x-15) = 180$, so $5x = 190$ (1); $x = 38$ (1); $\angle A = 58^\circ$, $\angle B = \angle C = 61^\circ$ (1).	1+1+1
22	Area $= \frac{5}{6} \times \frac{3}{10} = \frac{1}{4}$ m^2 (1); Smaller side squared $= \frac{9}{100}$ m^2 (1); Difference $= \frac{4}{25}$ m^2 (1).	1+1+1
23	For parallel lines these angles must be equal (1); Here 70° and 75° are not equal (1); The lines are not parallel (1).	1+1+1
24	$49 + 94 = 143$; $143 + 341 = 484$ (1); Number of steps $= 2$ (1); Palindrome: 484 (1).	1+1+1
	OR: Two sides and the included angle are equal, so SAS; $\triangle ABC \cong \triangle PQR$, hence $PR = AC = 12$ cm (1); Perimeter of PQR $= 6 + 9 + 12$ (1); $= 27$ cm (1).	1+1+1
25	Improper form $\frac{5}{3}$; $\frac{5}{3} \times \frac{6}{5}$ (1); $= \frac{30}{15}$ (1); Answer 2 (1).	1+1+1
	OR: The sum is 90° and the number of parts is 6 (1); One part $= 15^\circ$ (1); The angles are 15° and 75° (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Equal to 110°: the angle itself, its vertically opposite angle, its corresponding angle and its alternate interior angle, that is 4 angles (1); Every other angle forms a linear pair with one of these, so it measures 70° (4 angles) (2); Check: $110 + 70 = 180$ (2).	1+2+2
	OR: Reverse $= 549$ (1); Difference $= 945 - 549 = 396$ (1); $= 99 \times 4$ (1); In general $100a + c - 100c - a = 99(a-c)$ (1); So the difference is always a multiple of 99 (1).	1+1+1+1+1

27	$4x + 20 = 180$, so $x = 40$ (1); Angles 60°, 30°, 90° (1); Classification: right-angled (1); Exterior angles 120°, 150°, 90° (1); Their sum $= 360^\circ = 360^\circ$ (1).	1+1+1+1+1
	OR: First part $= \frac{1}{3} \times 90 = 30$ km (1); Remaining $= 90 - 30 = 60$ km (1); Second part $= \frac{3}{4} \times 60 = 45$ km (1); Left $= 60 - 45$ (1); $= 15$ km (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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